

The missing precision term: contradiction-blindness in APEX scoring

Research proposal for the Mercor APEX Fellowship. September 2026.

CLAIM

APEX-v1 scores a response as the fraction of binary rubric criteria a judge marks present. No criterion asks whether the response also asserts the opposite. That gap is exploitable, I've measured it, and it has a cheap fix. The project is to size it on the held-out set with real experts, test the fix, and report how much of the leaderboard spread survives.

WHAT I MEASURED

A pre-registered study on the 100 public APEX-v1 cases: 40 tasks, 10 per domain, hypotheses and falsifiers written before the data existed. GPT-4.1 wrote an honest response per task with attachments. I applied perturbations that make a response worse without removing correct content, then regraded with a reconstruction of the APEX judge (Gemini 3 Flash, one criterion per call). A second model family judged original vs perturbed blind, both orders, no rubric.

Perturbation of the honest response	Rubric score change	Blind judge prefers original
Append a section reversing every conclusion	0.000 (90% CI ± 0.026)	95%
Rewrite every claim as a hedge	0.000	85%
Add seven boilerplate sections	+0.002	90%
Make every fact wrong, same structure	-0.280	82%
Delete the second half	-0.177	100%
Regrade identical text (noise floor)	+0.004	n/a

The first three rows are the finding: responses a blind judge rates worse nine times in ten score the same. The fourth row is the failed prediction. I'd pre-registered "wrong content scores like truncation" as the falsifier for "the judge doesn't check correctness," and it triggered, so I'm not claiming that. The mechanism is narrower: wrongness that replaces the correct claim is caught, wrongness that sits beside it is free. The metric is recall of correct assertions and nothing else. A checkbox-stuffing attack built from other tasks' rubrics scored level with honest, but the blind judge couldn't tell it from honest either. Not evidence. Open.

WHY IT MATTERS

A model tuned on this metric learns to commit to an answer, then cover the alternatives. That response is worse for the client and scores the same. The leaderboard can't separate a model that knows the answer from one that lists the candidates. Any presence-only rubric benchmark has this property. APEX is where it can be measured because the rubrics and recipe are public.

PLAN

(1) Replace the LLM proxy with expert ratings: blind pairwise, committed vs hedged, four domains. (2) Run the suite against the real judge and prompt on the held-out 400, three generator families. (3) Build and test the fix: a negation pass in the judge (for each passed criterion, ask if the response also asserts its negation) and expert-written negative criteria. Measure cost, expert agreement, gap closed. (4) Rescore the leaderboard under the patched metric and report which rank gaps hold.

CAVEATS

One generator. Flash judge, not the paper's Pro ($\kappa = 0.92$ with Pro on a 54-criterion subset). LLM quality proxy. $n = 40$, one seed. Each is a reason to do the study properly, not a reason to doubt the direction.

Pre-registration, full results, and code: github.com/<your-handle>/apex-rubric-audit